Friction: Test the Surfaces · Knowledge Organiser

Learning goal: I can compare friction on different run-out surfaces using repeated measurements and a controlled setup.

Today I Can

I can keep the block, ramp height, and release point the same.

I can record repeated stopping distances in centimetres.

I can use the evidence to identify which surface gave greater resistance.

Must-Know Words

run-out surface: the flat test surface at the bottom of the ramp where the block slows and stops

stopping distance: how far the block travels along the run-out surface before stopping, in cm

control variable: a factor kept the same so the surface comparison is fair

repeat trial: running the same surface test more than once to check consistency

Core Science Steps

Keep the same block, low ramp, release mark, and no-push release. Change only the run-out surface.

Measure from the same zero mark at the bottom of the

ramp to the same front edge of the block in cm.

Repeat each surface twice and keep both results·shorter stopping distance shows greater resistance.

Keep the Science Clear

Repeated trials show whether the difference between surfaces is consistent rather than a one-off accident.

Our conclusion applies to this block and surface pair·do not claim it proves every material in the world.